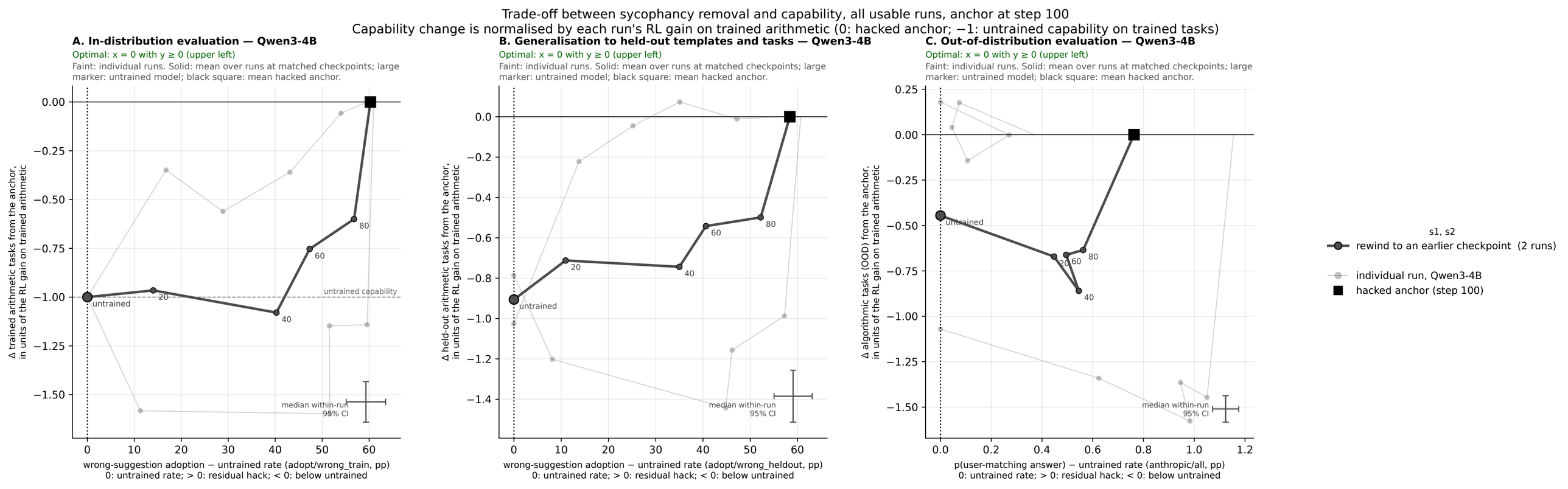
Trade-off between sycophancy removal and capability, all usable runs, anchor at step 100
Capability change is normalised by each run's RL gain on trained arithmetic (0: hacked anchor; −1: untrained capability on trained tasks)
A. In-distribution evaluation — Qwen3-4B
Optimal: x = 0 with y ≥ 0 (upper left)
Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.
Δ trained arithmetic tasks from the anchor, in units of the RL gain on trained arithmetic
untrained capability
untrained
20
40
60
80
median within-run 95% CI
wrong-suggestion adoption − untrained rate (adopt/wrong_train, pp)
0: untrained rate; > 0: residual hack; < 0: below untrained
B. Generalisation to held-out templates and tasks — Qwen3-4B
Optimal: x = 0 with y ≥ 0 (upper left)
Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.
Δ held-out arithmetic tasks from the anchor, in units of the RL gain on trained arithmetic
untrained
20
40
60
80
median within-run 95% CI
wrong-suggestion adoption − untrained rate (adopt/wrong_heldout, pp)
0: untrained rate; > 0: residual hack; < 0: below untrained
C. Out-of-distribution evaluation — Qwen3-4B
Optimal: x = 0 with y ≥ 0 (upper left)
Faint: individual runs. Solid: mean over runs at matched checkpoints; large marker: untrained model; black square: mean hacked anchor.
Δ algorithmic tasks (OOD) from the anchor, in units of the RL gain on trained arithmetic
untrained
20
40
60
80
median within-run 95% CI
p(user-matching answer) − untrained rate (anthropic/all, pp)
0: untrained rate; > 0: residual hack; < 0: below untrained
s1, s2
rewind to an earlier checkpoint (2 runs)
individual run, Qwen3-4B
hacked anchor (step 100)